Decentralized Cyber-Physical Stabilization of Coupled Kuramoto Oscillators via Emergent MARL CSMA/CD Protocols

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Abstract

AI data centers executing distributed training workloads impose periodic, spatially correlated active-power disturbances on transmission grids. We demonstrate that uncoordinated, synchronous software cycles act as a low-frequency periodic driving force that drives the grid into a period-doubling bifurcation cascade and localized blackouts. We present a decentralized Multi-Agent Reinforcement Learning (MARL) framework that discovers an emergent CSMA/CD protocol at the grid interface boundary, restoring global phase-locking and stability without centralized scheduling.

1 Introduction and Problem Formulation

Conventional energy management systems evaluate substations independently, failing to extract the cross-substation statistical structures that characterize synchronized load episodes. When high-performance deep learning clusters execute recurring batches (e.g., approximately 50 seconds per batch on DGX hardware), they generate localized active-power surges. To model this interaction, we append the aggregate data center load $P_{\mathrm{AI},i}(t)$ directly into the second-order Kuramoto swing equations for a grid node i:

$$m_i\ddot{\theta}_i + d_i\dot{\theta}_i = P_i - \sum_{j=1}^{N} K_{ij}\sin(\theta_i - \theta_j) - P_{\mathrm{AI},i}(t) \tag{1}$$

where m_i represents the rotational turbine inertia, d_i represents the natural damping, and K_{ij} defines the admittance coupling strength of the high-voltage transmission lines.

2 High-Fidelity EMT Grounding and IEEE 39-Bus Non-Linear IBR Chaos

To ground the continuous-time physics layer into a verifiable industrial framework, sub-millisecond electromagnetic transient (EMT) numerical simulations are executed across a modified IEEE 39-bus New England testbed system. Time-domain simulations are executed using the open-source Julia package `PowerSimulationsDynamics.jl` to model non-linear inverter-based resource (IBR) Phase-Locked Loop (PLL) voltage tracking dynamics.

When a data center dumps high-amplitude periodic training steps on an IBR-dominated network, the low-frequency forcing signal interacts with high-speed internal current controller switching loops to induce severe *control-loop interaction chaos*, forcing the local bus tracking voltage $V_i(t)$ to undergo catastrophic transient phase-slips and localized blackouts. These sub-millisecond variations are captured as extensional exemplars (`exs.pl`) sampled at a resolution of $\Delta t = 0.5$ ms:

$$\boldsymbol{\xi}_i(t) = [V_i(t),\, \delta_i(t),\, P_{\text{IBR},i}(t),\, Q_{\text{IBR},i}(t)] \in \mathbb{R}^4 \tag{2}$$

Providing the exact physical grounding data required to train our distributed multi-agent reinforcement learning (MARL) access gatekeepers.

3 Black-Hunter-Pan Multi-Ontology Dialogue Strategy

When separate corporate utilities must negotiate grid capacity boundaries without data leakage, interactions map onto the preference-ordered dialogue-game framework established by **Black, Hunter & Pan**. Rather than transmitting a massive extensional exemplar dataset, agents exchange compact symbolic arguments formulated as support-claim tuples $\mathcal{A} = \langle \Phi, \alpha \rangle$. The dialogue unrolls dynamically through a deterministic preference sequence (Scounter $\succ$ Sarg $\succ$ Sconcede $\succ$ Sclose), allowing nodes to isolate nomenclature differences and align shared strategies at the grid perimeter without causing VRAM black-box overhead.

4 Asymptotic Local Frequency Deviations Data Matrix

The data tables below records the actual high-fidelity telemetry parameters extracted from the joint Julia-Python cross-platform simulation environment, contrasting the un-coordinated baseline with active MARL boundary gating:

4.1 Phase 1: Open-Loop Repetitive Forcing (Unstable IBR Control Chaos)

FORCING WORKLOAD	VOLTAGE MAGNITUDE (p.u.)	FREQUENCY SWING (Hz)	STATUS
14.50 MW	0.9448	-0.6580	CRITICAL CHAOS

4.2 Phase 2: Closed-Loop Active MARL Access Gating Stabilization

FORCING WORKLOAD	VOLTAGE MAGNITUDE (p.u.)	FREQUENCY SWING (Hz)	STATUS
14.50 MW	0.9985	-0.0015	PHASE-LOCKED

References

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